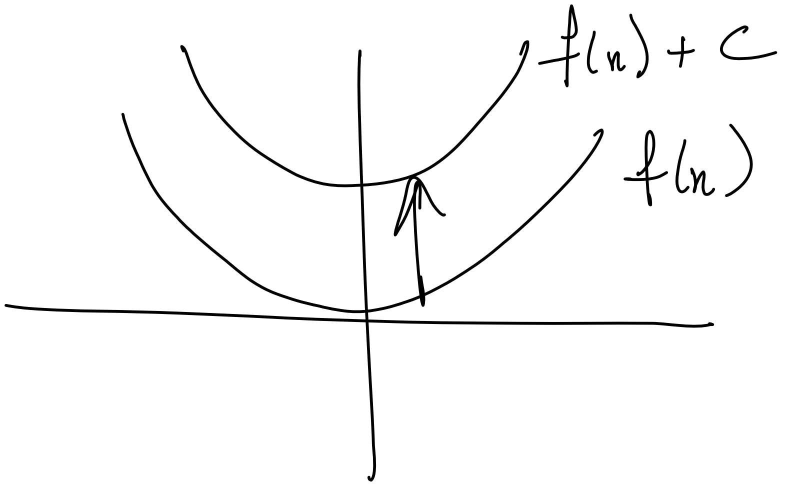
Note Title 2015/02/08

Unit 1.6 New functions from old p 36-44

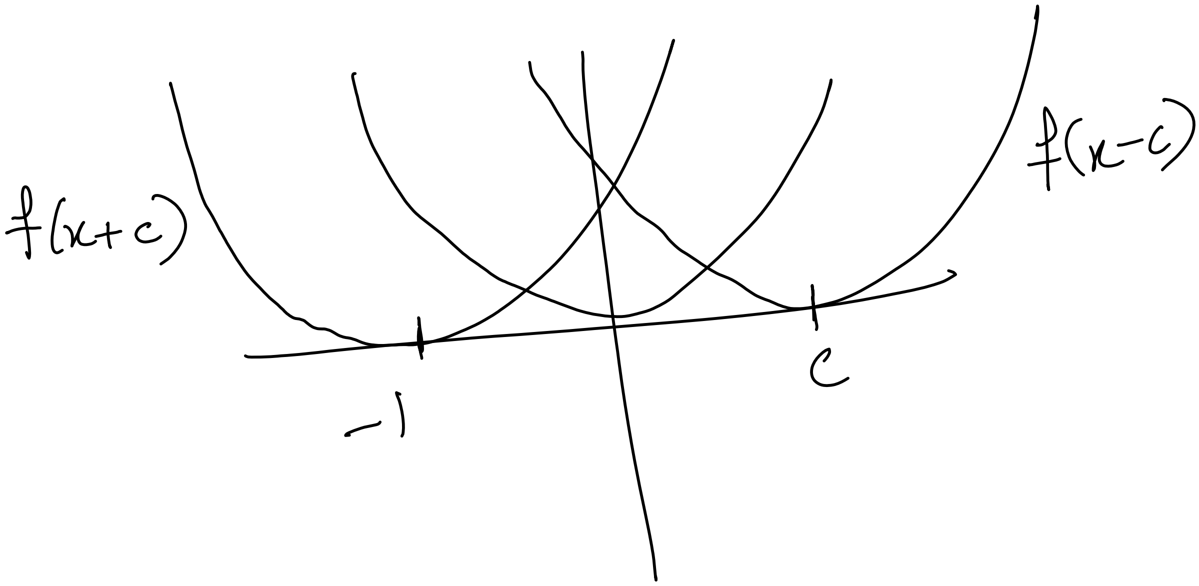
1. Vertical and horizontal shifts $(c > 0)$

$f(x) + c$ ↑

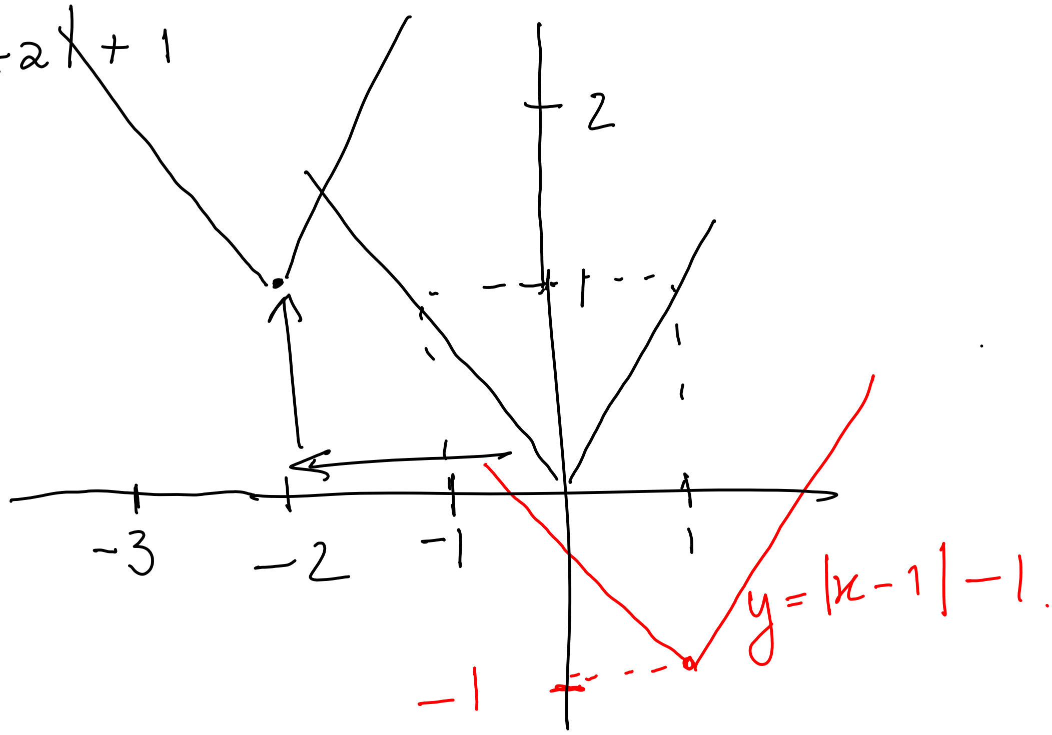
$f(x) - c$ ↓



$c > 0$



Eg $f(x) = |x+2| + 1$



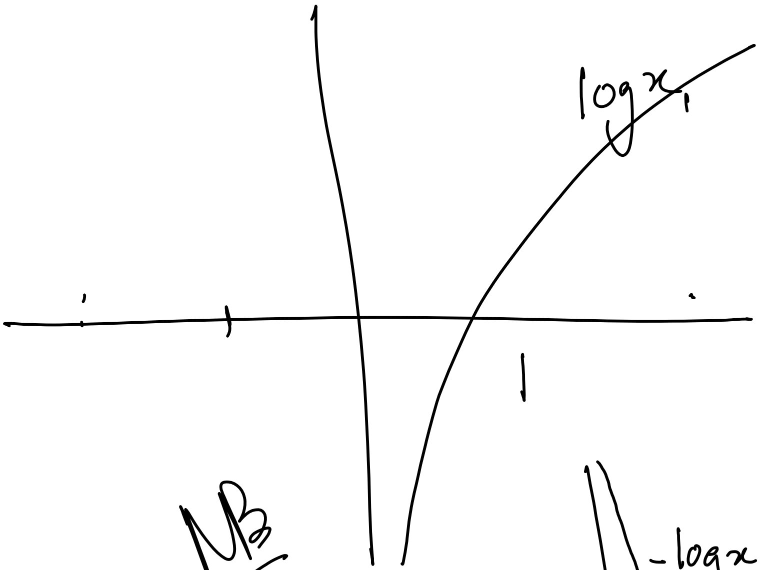
2. Reflections

$f(-x) \to$ y-axis reflection

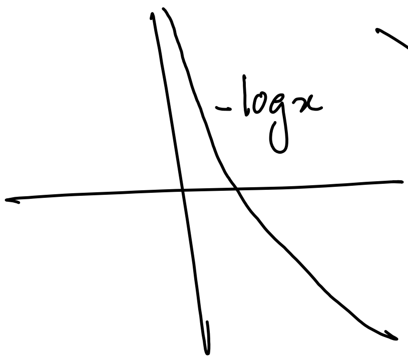
$-f(x) \to$ x-axis reflection

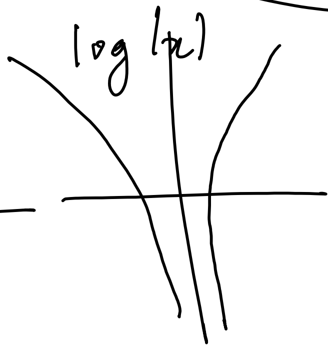
Eg. $y = \log x$, $y = -\log x$, $y = \log(-x)$, $y = \log|x|$.

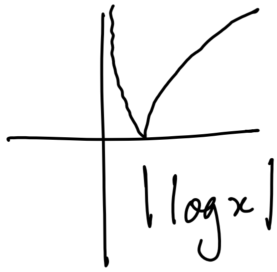
$$\begin{cases} \log x & \text{if } x > 0 \\ \log(-x) & \text{if } x < 0 \end{cases}$$



NB



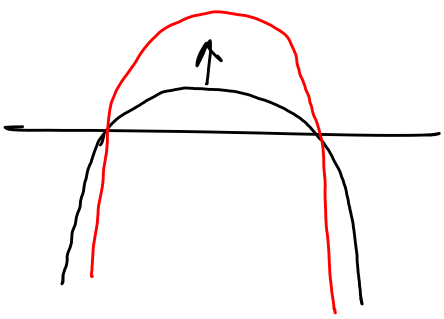




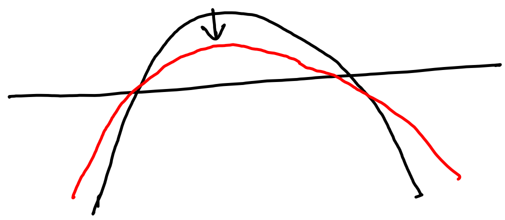
3. Stretching and compressing

$a f(x)$, $a > 0$

$a > 1$



$a < 1$



4. Sum / difference / product / quotient of functions

$f(x) = x+1$, $g(x) = x^2$

$f(x) + g(x) = x+1+x^2$

$f(x) \cdot g(x) = (x+1)x^2$

$\dfrac{f(x)}{g(x)} = \dfrac{x+1}{x^2}$, $x \neq 0$

5. Composite functions $\underline{\underline{\text{NB}}}$

$f(g(x))$ also $f \circ g(x)$

$\underline{f(x) = x^2}$, $g(x) = x+1.$

$f(g(x)) = f(x+1) = (x+1)^2$

$g \circ f(x) = g(f(x)) = g(x^2) = x^2+1$

$f \circ f(x) = f(f(x)) = f(x^2) = x^4$

$$f \circ g \circ g(x) = f(g(g(x)))$$
$$= f(g(x+1))$$
$$= f(x+2)$$
$$= (x+2)^2$$

Al-jabr

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al shalan
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Descartes ~ 1600

$f \circ g(x)$
$= f(x) \times g(x)$